Data Foundations Training

Product 1: Solar Panels

Sample data: A solar panel company sold 400 units in January and 520 units in February.

1. Descriptive Analysis – What happened?

Sales increased from 400 to 520 solar panels, representing a 30% increase in monthly sales.

2. Diagnostic Analysis – Why did it happen?

The increase may be associated with government incentives, rising electricity costs, or successful marketing campaigns. Sales records and campaign data should be reviewed to identify the actual cause.

3. Predictive Analysis – What might happen next?

If demand continues to grow, solar panel sales may increase in the next month. Future demand will depend on weather, installation capacity, prices, and incentive programs.

4. Prescriptive Analysis – What should we do?

The company should maintain sufficient inventory, train more installation teams, offer suitable financing options, and monitor monthly demand.

Product 2: 3D Printers

Sample data: A retailer received 80 customer orders in March and 60 in April.

1. Descriptive Analysis – What happened?

Orders decreased by 20, representing a 25% decline from March to April.

2. Diagnostic Analysis – Why did it happen?

Possible reasons include high product prices, limited material compatibility, delivery delays, or reduced advertising. Order cancellations, customer feedback, and pricing records should be analyzed to confirm the cause.

3. Predictive Analysis – What might happen next?

If the decline continues, the retailer may receive fewer orders in the following month. A reliable forecast should consider past demand, new product launches, and customer interest.

4. Prescriptive Analysis – What should we do?

The retailer should review customer complaints, demonstrate printer capabilities, offer beginner bundles, and test targeted promotions to improve sales.

Product 3: Electric Vehicle Charging Stations

Sample data: A charging station recorded 1,500 charging sessions in June and 1,950 in July.

1. Descriptive Analysis – What happened?

Charging sessions increased by 450, representing a 30% increase from June to July.

2. Diagnostic Analysis – Why did it happen?

The increase may be due to more electric vehicles in the area, improved station visibility, or additional charging points. Vehicle traffic, station availability, and charging records should be analyzed to identify the main cause.

3. Predictive Analysis – What might happen next?

Charging demand may continue to increase if electric vehicle usage grows. Future demand will also depend on charging prices, competing stations, and local travel patterns.

4. Prescriptive Analysis – What should we do?

The operator should monitor peak usage, maintain charging equipment, plan capacity upgrades where needed, and use demand forecasts to schedule maintenance during quieter periods.

Product 4: Air Purifiers

Sample data: A company sold 800 air purifiers in May and 1,000 in June.

1. Descriptive Analysis – What happened?

Air purifier sales increased by 200 units, representing a 25% increase from May to June.

2. Diagnostic Analysis – Why did it happen?

The increase may be associated with rising concerns about air quality, seasonal pollution, or successful promotional campaigns. Sales records, air-quality trends, and marketing data should be analyzed to identify the actual cause.

3. Predictive Analysis – What might happen next?

Sales may continue to increase if air-quality concerns remain high. However, future demand will depend on pollution levels, seasonal changes, prices, and competitor activity.

4. Prescriptive Analysis – What should we do?

The company should maintain sufficient stock, promote suitable air purifier models, offer filter replacement packages, and use sales forecasts to plan inventory and marketing activities.

Product 5: Water Purifiers

Sample data: A company completed 600 water purifier service requests in August and 750 in September.

1. Descriptive Analysis – What happened?

Service requests increased by 150, representing a 25% increase from August to September.

2. Diagnostic Analysis – Why did it happen?

The increase may be associated with seasonal water-quality concerns, more installed units, filter replacement cycles, or equipment issues. Service categories, installed-unit counts, and maintenance histories should be reviewed to identify the reason.

3. Predictive Analysis – What might happen next?

Service demand may remain high if many customers require scheduled filter replacements or maintenance. Forecasts should account for installed units, service intervals, and seasonal patterns.

4. Prescriptive Analysis – What should we do?

The company should forecast spare-filter requirements, schedule technicians efficiently, remind customers about preventive maintenance, and investigate recurring product faults.

Conclusion

These five products demonstrate how data analysis can support business decisions:

- Descriptive analysis: Summarizes what happened using historical data.
- Diagnostic analysis: Investigates why an outcome occurred.
- Predictive analysis: Estimates possible future outcomes.
- Prescriptive analysis: Recommends actions to improve results.

Combining these approaches helps organizations improve sales, customer satisfaction, product reliability, service planning, and operational efficiency.